

Earth's Curvature

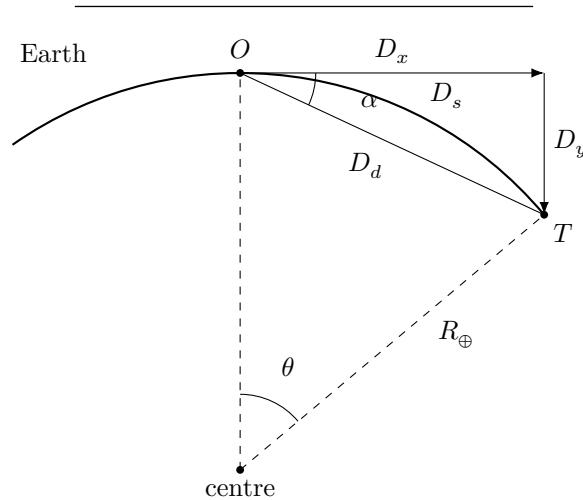
Jamie Breault

This document derives the geometry used to place a sampled point into the two-dimensional vertical cross-section that line-of-sight analysis works in (see `LineOfSight.process_full_line_of_sight` in `scripts/commons.py`).

A point sits at surface distance D_s from the observer (measured along the surface of the Earth at sea level) and at elevation h above sea level. We want its position relative to the observer's horizontal plane.

Notation

- $R_\oplus$ — mean radius of the Earth at sea level.
- D_s — surface distance between the observer and the point.
- θ — central angle between the observer and the point, $\theta = D_s/R_\oplus$.
- D_x — horizontal distance from the observer to the point's projection on the observer's horizontal plane.
- D_y — vertical distance of the point above the observer's horizontal plane.
- α — angle between the observer's horizontal plane and the straight line to the point.



Geometry of a point at central angle θ from the observer O , relative to O 's

horizontal plane.

Horizontal distance:

$$\sin(\theta) = D_x/R_\oplus \implies D_x = R_\oplus \sin\left(\frac{D_s}{R_\oplus}\right)$$

Vertical drop due to curvature:

$$R_\oplus = R_\oplus \cos(\theta) + D_y \implies D_y = R_\oplus \left(1 - \cos\left(\frac{D_s}{R_\oplus}\right)\right)$$

Angle below the horizontal plane:

$$\tan(\alpha) = \frac{D_y}{D_x} = \frac{1 - \cos(D_s/R_\oplus)}{\sin(D_s/R_\oplus)} = \tan\left(\frac{D_s}{2R_\oplus}\right)$$

$$\alpha = \frac{D_s}{2R_\oplus}, \quad 0 < \alpha < \pi/2$$

A point at elevation h lies at radius $R_\oplus + h$, so its Cartesian position (observer at the origin) is

$$x = (R_\oplus + h) \sin \theta, \quad y = (R_\oplus + h) \cos \theta - R_\oplus - h_1,$$

where h_1 is the observer's elevation. The cross-section is then rotated by α so the straight observer-target line lies on the x-axis, making ground heights directly comparable to the light-ray heights derived in `atmospheric_refraction.md`.